



Media - films and games

Task A: Films

1) Investigation time.

You will need:

- two rulers (same size)
- one tape measure

Place your ruler on the shorter edge on a table (you may need to prop it against some books). Place the other ruler further away in the same orientation.

Sit so that your eyes are level with top of the 'smaller' ruler. Use the 'larger' ruler to estimate how tall the 'smaller' ruler is.

Now measure the distance between where you sat and each ruler using the tape measure.

What is the relationship between the two distances and the perceived height of the 'smaller' ruler?

Repeat this investigation multiple times, placing the rulers at different points.

What do you notice?

The 'larger' ruler was 10 cm away and the 'smaller' ruler was 15 cm away.

The 'larger' ruler was two thirds of the distance of the 'smaller' ruler.

The 'smaller' ruler was two thirds of the height of the 'larger' ruler.

Task B: Games

Objective - reach the goal in 3 turns.

You are player A.

Write down the translation required to get to the goal.

If you would move to the same space as the computer, you end your movement (unless you reach the goal). Movement is in one unit increments.

You must make it past the computer and to the goal in 3 turns or less.

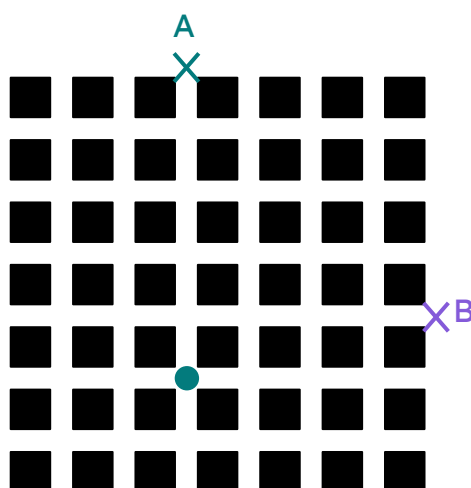
1) ● = Goal

Player A

E.g.

Round 1: 1 unit left,
5 units down

Round 2: 1 unit
right



Computer (B)

Round 1: 4 units left

Round 2: 1 unit right, 1 unit
down

Round 3: 2 units up

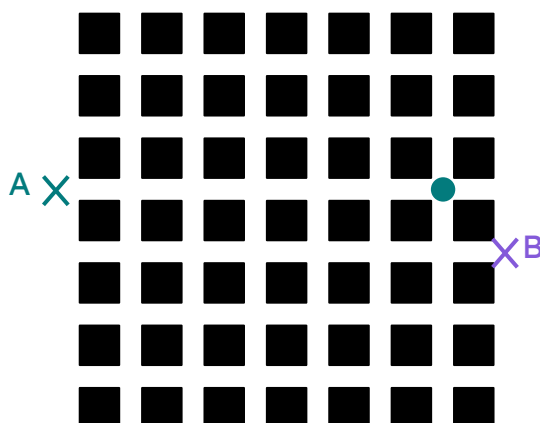
2) ● = Goal

Player A

E.g.

Round 1: 1 unit up,
6 units right

Round 2: 1 unit
down



Computer (B)

Round 1: 2 units left, 1 unit
up

Round 2: 1 unit left

Round 3: 1 unit up,
2 units right



3) Design your own game. Decide where A and B will start and where the goal will be. Write your instructions for the computer. Swap with a peer and see if you can beat each other's game.

You could modify the game by allowing more complicated rules should you wish.

